

DN SC 6330: Responsible Machine Learning*

Lecture 07: Responsible AI Management and Global Regulatory Frameworks

Michael Akinwumi

Adjunct Professor of Responsible Machine Learning

Department of Decision Sciences

The George Washington University, School of Business

April 27 & 28, 2026



*This material is shared under a [CC BY 4.0 license](#)  which allows for editing and redistribution, even for commercial purposes. However, any derivative work should attribute the author.

Outline (2.5 hours)

- 1 Lecture Overview & Learning Objectives
- 2 Part I: Post-Deployment Responsibilities
- 3 Part II: Global AI Governance Frameworks
- 4 Part III: Stakeholder & Regulatory Gap Analysis

What This Lecture Covers

Post-Deployment Responsibilities

- Data cards, model cards, and system cards
- Continuous evaluation and monitoring
- Alerting and rollback mechanisms
- Human-in-the-loop design
- Organizational roles for deployment

Governance & Regulatory Frameworks

- System-level AI risks in ML models
- NIST AI Risk Management Framework
- EU AI Act
- ISO/IEC 42001 and ISO/IEC 23894
- Three Lines of Responsibility

What This Lecture Covers

- Regulatory gap analysis

Session Structure

Part I: Post-Deployment Responsibilities (45 min) → Break (5 min) →
Part II: Governance Frameworks (45 min) → Break (5 min) → Part III:
Stakeholder & Gap Analysis (30 min)

Learning Objectives

By the end of this lecture, you will be able to:

- 1 Describe the purpose and components of data cards, model cards, and system cards as post-deployment artifacts.
- 2 Explain what continuous evaluation, alerting, and rollback mechanisms look like in an operational ML context.
- 3 Identify the organizational roles required for responsible AI deployment and monitoring.
- 4 Characterize system-level AI risks associated with ML models in production.

Lecture Recap — Where We Have Been

Previous Lectures

Lecture	Core Topic
01	Alignment and Foundations (COMPAS)
02	Explainability: LIME, SHAP, DiCE
03	Algorithmic Bias Measurement: AIR, ME, SMD
04	Robustness, Generalization, Dataset Drift (PSI, KS, ADWIN)
05	ML Security and Abuse Pathways
06	Bridging to the Capstone
07	Responsible AI Management & Global Frameworks

Lectures 01–06 focused on *technical* dimensions of responsible ML. Today we zoom out to the *organizational and regulatory* layer that makes those technical practices stick.

Why Post-Deployment Matters

The Cambridge Analytica Lesson

Cambridge Analytica used predictive AI to infer personality types from Facebook data and targeted political ads at users subconsciously. The consequences: a \$5 billion fine for Facebook and the **destruction of Cambridge Analytica itself**. Its irresponsible use of AI ended the company.

- Model deployment is not the finish line; it is the start of **ongoing responsibility**.
- AI risks – bias, drift, hallucination, manipulation, privacy invasion – do not disappear after launch.
- **Responsible AI (RAI)** is defined as AI that aligns with societal values – an ongoing, not one-time, commitment.

System-Level AI Risks in Production

Individual Model Risks

- Privacy invasion via inference
- Harmful bias and unfairness
- Hallucination (false outputs that appear true)
- Manipulation of user behavior
- Intellectual property violation
- Misapplication beyond intended scope

Aggregate (System) Risks

- Interactions among interconnected models
- Reliance on common assumptions or shared data
- Correlated failures across multiple models simultaneously
- Feedback loops amplifying errors at scale
- Misinformation and hate speech at scale

System-Level AI Risks in Production

Key Principle

Sound practice involves assessing model risk **both individually and in aggregate**.

Aggregate risk reflects interactions and dependencies that can adversely affect several models and their outputs simultaneously.

Documenting What You Deploy: Data Cards

Data Cards

Structured documentation artifacts that describe the dataset used to train, validate, or test a model—including provenance, composition, collection methods, known limitations, and recommended uses.

Key dimensions a data card addresses:

- **Provenance:** Where did the data come from? Who collected it and how?
- **Composition:** What subgroups, features, and time periods are represented?
- **Known biases:** What demographic or historical skews exist in the data?
- **Limitations:** What populations or conditions are underrepresented?

Documenting What You Deploy: Data Cards

- **Intended use:** For what tasks was this data appropriate?

Example from the course: The COMPAS recidivism dataset requires a data card disclosing its racial composition imbalances – the same imbalances that drove the ProPublica 2016 audit findings we studied in Lecture 04.

Documenting What You Deploy: Model Cards

Model Cards

Standardized documentation artifacts that accompany a trained model and communicate its intended uses, performance characteristics, evaluation results across subgroups, and known limitations.

A model card should include:

- Model type and architecture
- Intended use cases and contexts
- Evaluation datasets and metrics
- Disaggregated performance by subgroup
- Known failure modes
- Ethical considerations

Documenting What You Deploy: System Cards

System Cards

Documentation that describes an entire AI system—not just a single model—including how multiple components interact, what data flows between them, and what guardrails govern the system as a whole.

System cards go beyond individual model cards by capturing:

- The composition of multiple models or decision-making components
- Human-in-the-loop touchpoints and override mechanisms
- Feedback loops that can amplify or correct errors
- Stakeholder roles and accountability chains
- System-level risks that emerge only from component interactions

Documenting What You Deploy: System Cards

Key Distinction

- ① Data Card → what went *in*.
- ② Model Card → what the *model* does.
- ③ System Card → what the *system* does in context.

Human-in-the-Loop (HITL) Design

Human-in-the-Loop

A design pattern in which a human decision-maker is kept in the decision process – reviewing, approving, or overriding model outputs – rather than allowing the model to act fully autonomously.

Why HITL is required:

- Machines can be optimized for a given definition of fairness, but **humans must decide when to do so and which definition to apply**
- High-stakes decisions affecting individuals require human judgment
- Prevents fully autonomous systems from causing irreversible harm

Human-in-the-Loop (HITL) Design

HITL touchpoints in a deployment:

- ① Model output review before consequential action
- ② Override capability for front-line users
- ③ Escalation path for ambiguous or high-stakes cases
- ④ Periodic governance review

Human-in-the-Loop (HITL) Design

The Three Lines of Responsibility

Line	Who	Deployment Responsibilities
First Line	Front-line data science teams; model users	Implement RAIM policies; spot risks; escalate issues; monitor controls
Second Line	RAI Officer; RAI Working Group; RAI Executive Committee	Formulate and monitor policies; decide grey-area issues; track regulatory developments
Third Line	Internal Audit	Evaluate whether policies and controls are rigorous, effective, and properly implemented

Vendor and Third-Party Model Governance

The Vendor Problem

When an organization uses third-party or vendor models, they may not receive the underlying code, data, or methodology because it is proprietary. Yet **post-deployment responsibilities remain fully applicable**.

Sound practice for vendor models:

- Develop an understanding of conceptual soundness, design, and development data
- Conduct ongoing monitoring and outcomes analysis to assess fit-for-purpose status
- Where vendor models are customized, **document, justify, and evaluate all adjustments**

Vendor and Third-Party Model Governance

- Maintain internal oversight and integrate vendor work into broader model risk management
- Clearly define roles and responsibilities when delegating validation activities

5-Minute Break

We resume with Part II: Global Governance Frameworks

Why Organizations Need Governance Frameworks

The Gap Problem

- Existing laws (privacy, anti-discrimination) leave many important AI gaps
- AI-specific laws impose management requirements but **draw few substantive lines**
- Organizations must figure out how to establish these boundaries themselves
- International standards help fill the vacuum

Regulatory Landscape

- EU AI Act (enacted 2024)
- Nearly 700 AI-related bills introduced in U.S. state legislatures in 2026 to date
- 31 U.S. states passed AI-related legislation or resolutions in 2024

Why Organizations Need Governance Frameworks

- Growing AI litigation in U.S. courts

The Management Opportunity

Responsible AI management prepares organizations to meet **emerging and future legal standards** and so reduce their legal and regulatory risk.

The Four Pillars of Responsible AI Management

1. Structure

Accountable people and committees in the right positions

2. Process

Systematic spotting, escalation, resolution, documentation

3. Substantive Standards

Lines between acceptable and unacceptable AI practices

4. Culture

Organizational values that fill gaps when policy is unclear

All four pillars are addressed within frameworks like NIST AI RMF, EU AI Act, and ISO/IEC standards.

The NIST AI Risk Management Framework

What It Is

Published by the National Institute of Standards and Technology (2023), the AI RMF provides a voluntary framework to help organizations manage AI risks more effectively throughout the AI lifecycle.

Core Functions (GOVERN – MAP – MEASURE – MANAGE)

- **GOVERN:** Culture, policies, roles, accountability
- **MAP:** Context and risk identification
- **MEASURE:** Analysis, assessment, benchmarking
- **MANAGE:** Prioritize, respond, monitor risks

What NIST Addresses

- Provides a model for policies and controls

The NIST AI Risk Management Framework

- Defines a fully fleshed-out AI governance program
- Can serve as a standard for **third-line audit activities**
- Accompanied by an AI RMF Playbook with actionable guidance

Scope

Applies to the full AI lifecycle—design, development, deployment, and de-commission.

The EU AI Act (2024)

What It Is

The European Union Artificial Intelligence Act is the world's first horizontal regulation on AI, enacted June 2024. It imposes a **risk-based, tiered regulatory regime** on AI systems placed on the EU market.

Risk Tiers

- ❶ **Unacceptable risk:** Prohibited (e.g., social scoring, subliminal manipulation causing significant harm)
- ❷ **High risk:** Strict conformity requirements (biometrics, critical infrastructure, employment, credit, education)
- ❸ **Limited risk:** Transparency obligations
- ❹ **Minimal risk:** No mandatory requirements

Key Obligations for High-Risk AI

The EU AI Act (2024)

- Risk management system throughout the lifecycle
- Data governance and quality controls
- Technical documentation and record-keeping
- Transparency and provision of information to users
- Human oversight measures
- Accuracy, robustness, and cybersecurity requirements

EU AI Act: Substantive Line-Drawing in Practice

What the EU AI Act Expressly Prohibits

Placing into the market an AI system that **deploys subliminal techniques beyond a person's consciousness** so as to induce that person to make a decision that causes them **significant harm**.

The EU AI Act (2024)

Relevance to Cambridge Analytica:

Cambridge Analytica's use of predictive AI to send ads subconsciously appealing to inferred personality types would be *directly targeted* by this prohibition.

Why the EU AI Act matters even for U.S. organizations:

- Extra-territorial reach: applies to AI systems with effects in the EU
- Sets the **direction in which law and public norms are headed** globally
- Organizations that follow its spirit are better prepared for future U.S. regulation
- Provides intelligence for substantive line-drawing even where not legally required

Laws Governing AI

General Laws Applied to AI

- Privacy law: gives individuals control over PII—but predictive AI *infers* sensitive information from innocuous data, creating an end-run around protections
- Anti-discrimination law: addresses *allocational* harms—but says little about quality-of-service or representational harms
- Both leave important gaps when applied to AI

AI-Specific U.S. Laws

- Colorado Privacy Act (2021): requires algorithmic impact assessments
- Colorado AI Act (2024): defines “consequential” AI decisions
- Algorithmic Accountability Act (2023, proposed): impact assessment obligations
- 31 states passed AI-related legislation in 2024 alone

Laws Governing AI

The Gap: New and proposed U.S. AI laws principally impose **management requirements**—they do not draw the lines between permissible and impermissible AI practices.

Laws Governing AI

Six strategies organizations can use to build substantive standards:

- 1 **Draw from corporate values:** Start with the mission statement and values—but these often require further operationalization.
- 2 **Follow the spirit of the law:** Apply the underlying policy intent of regulations, not just the letter—and anticipate where law is heading.
- 3 **Benchmark against peer organizations:** Useful for staying competitive, but does not allow differentiation on trustworthiness.
- 4 **Scan the environment for emerging standards:** Track law, regulation, and public opinion trends (e.g., environmental scanning).
- 5 **Develop and use precedent decisions:** Capture RAI Committee decisions and use them as binding precedents for future cases.
- 6 **Adopt AI ethics principles:** Use them to sensitize staff and spot issues—combined with the above strategies.

5-Minute Break

We resume with Part III: Stakeholder & Regulatory Gap Analysis

Responsible AI Management Creates Business Value

The Ohio State University 2023 survey of corporate AI governance managers found: **Overall value from RAIM investment**

- 68% said RAI was “important” or “very important”
- Nearly 40% said it created “a lot” or a “great deal” of value
- Most others reported “moderate” value
- Only a small fraction reported no value

Types of value (ranked by average score)

- ① Product quality improvement
- ② Enhanced competitive advantage
- ③ Reputation and trust
- ④ Reduced regulatory and legal risk
- ⑤ Achievement of corporate values
- ⑥ Enhanced employee relations

Responsible AI Management Creates Business Value

The Core Message

Responsible AI is not just the right thing to do; it is **essential to business success in the algorithmic economy**.

Spotting Issues: Environmental Scanning

Environmental Scan

The acquisition and use of information about events, trends, and relationships in an organization's external environment, the knowledge of which would assist management in planning the organization's future course of action.

Five steps of an AI-focused environmental scan:

Responsible AI Management Creates Business Value

- ① **Define scope:** AI risks (privacy, bias, hallucination), responsible AI opportunities, using the PEST framework (Political, Economic, Social, Technological)
- ② **Assign responsibility:** Typically the RAI Officer or their team; multiple perspectives are valuable
- ③ **Scan the environment:** News media, government reports, academic studies, thought-leader blogs, legal trackers
- ④ **Summarize and analyze:** Concise highlights, core trends, opposing views, citations
- ⑤ **Repeat:** Quarterly scans recommended during this period of rapid AI change

Risk and Opportunity Sweeping

A cross-functional team applies the environmental scan framework to the organization's actual AI activities. The team asks—for each stakeholder affected:

- How will this AI activity impact stakeholders? Benefit or harm them?
- Does it pose any legal compliance or liability risks?
- Will it generate significant controversy or public concern?
- Is it consistent with the company's mission and corporate values?
- Are there opportunities to achieve advantage through more responsible AI?

Risk and Opportunity Sweeping

Risk classification matrix:

	Probability	
	Low	High
Magnitude		
Low	Low	Med
High	Med	High
Very High	High	Very High

Organizations use this ranking to **prioritize corrective actions** and resource allocation.

Building a Responsible AI Culture

Why Culture Cannot Be Skipped: All accountability systems have blind spots. Culture shapes what employees do **when no one is looking**. A robust RAI culture fills the gaps in law and policy that even the best management structures cannot fully address.

Key culture-building practices:

- State RAI goals clearly from the top (C-suite endorsement)
- Measure and incentivize RAI performance (KPIs, bonuses)
- Appoint responsible AI ambassadors / champions
- Conduct RAI post-mortems on ethics incidents
- Think through real-world RAI scenarios with no clear answers
- Bring in external speakers including those harmed by AI

Building a Responsible AI Culture

- Hire for responsible AI (interview for ethics reasoning)
- Hold RAI-themed retreats and interest groups
- Explain the upside: trust, reputation, competitive advantage
- Remind all employees they are subject to AI decisions too

Key principle: Culture-building must be grounded in a **system**, not a single charismatic individual. Individuals leave; systems endure.

Grading and Policy

- Grading
 - Weekly assignment: $\frac{2.5}{10}$
 - Public GitHub with working codebase: $\frac{2.5}{10}$
 - Weekly quiz: $\frac{2}{10}$
 - Capstone: $\frac{3}{10}$
- Coding/Scripting
 - Python
 - R
- Collaborative Infrastructure
 - GitHub

Grading and Policy

- Capstone project
 - 2024 Home Mortgage Disclosure Act (HMDA): Loan/Application Records (LAR) [data](#) 
 - Public LAR data: [dictionary](#) 
 - Goal: Develop a classification model that predicts `action_taken` and apply principles taught in class to the model.
 - Label creation: `action_taken` values 1 and 2 coded as 1 and 3 coded as 0 while records matching other values are filtered out.
 - Group (no more than 5 members and no less than 3 members)
 - Group members are randomly assigned by instructor using the standard *math* and *random* libraries in **Python**.
- Syllabus: [link](#) 
- Officer hours: Fungler Hall 415R, Fri, 5:00PM - 7:00PM
- Class resources:
 - [Instructor's materials](#) 
 - [Additional resources](#)  from Patrick Hall